

Last week : Poisson bracket, Scattering

$AdS_5 \times S^5$ superstring

This week : $SL(2)$ sector

large spin limit, transcendentality

BFKL limit

Asymptotic Bethe Ansatz

[Beisert Staudacher, 0307042; Beisert, 0511082]

§ $SL(2)$ sector

$N=4$ SYM has three rank-1 subsectors

$$su(2) : \text{tr}(Z^J W^{J^2}) + \text{perm}, \quad s = 1/2 \quad XXX$$

$$su(1|1) : \text{tr}(Z^J \psi^M) + \text{perm}, \quad S_{12} = -1 \quad (\text{free})$$

$$\underline{sl(2)} : \text{tr}(D_+^S Z^J) + \text{perm}, \quad s = -1/2 \quad XXX$$

has lot of data to test Bethe Ansatz & AdS/CFT

Cusp dimensions, (conformal) Regge limit, transcendentality...

One-loop Bethe Ansatz is determined by
rationality + global symmetry ($PSU(2,2|4)$)

Bethe Ansatz for spin s XXX model:

$$\left(\frac{u_j + i/2}{u_j - i/2} \right)^L = \prod_{k \neq j}^M \frac{u_j - u_k + 2is}{u_j - u_k - 2is}$$

Anomalous dimensions : $\gamma = \frac{1}{8\pi^2} \sum_{j=1}^M \frac{1}{u_j^2 + 1/4}$

$S \leq 0$: non-compact spin chain

How to obtain Hamiltonian?

$$\mathcal{Q} = \text{tr} (D_+^S Z^T) + \dots$$

$$= \text{tr} ([n_+^\mu D_\mu]^S Z^T) + \dots$$

$$n_+^\mu = (n^+, n^-, n^2, n^3)$$

$$= (1, 0, 0, 0)$$

: null vector

naively

$$\langle \partial_+^S \bar{Z}(x) \partial_+^S Z(y) \rangle \sim \left(\frac{\partial}{\partial x^+} \right)^S \left(\frac{\partial}{\partial y^+} \right)^S \frac{1}{|x-y|^2}$$

$$\sim \frac{(x_+ - y_+)^{2S}}{|x-y|^{2+2S}}$$

CFT 2-point fn with spin

$$\langle \mathcal{O}_{\mu_1 \dots \mu_s}(x) \mathcal{O}_{\nu_1 \dots \nu_s}(y) \rangle = \frac{I_{\mu\nu}^{\rho, \vec{\nu}}(x-y)}{|x-y|^{2\Delta}}$$

$$I_{\mu\nu}^{\rho, \vec{\nu}}(x) = \text{Sym} \left\{ \prod_{k=1}^s \left(\eta_{\mu_k \nu_k} - \frac{2x_{\mu_k} x_{\nu_k}}{x^2} \right) \right\}$$

Symmetric traceless part

$$\text{If } \mathcal{O}_+ = n_+^{\mu_1} \dots n_+^{\mu_s} \mathcal{O}_{\mu_1 \dots \mu_s}, \quad n_+^\mu n_+^\nu I_{\mu\nu} = \frac{-2(x_+)^2}{x^2}$$

$$\langle \mathcal{O}_+(x) \mathcal{O}_+(0) \rangle \sim \frac{(x_+)^{2s}}{|x|^{2\Delta+2s}}$$

$\simeq$ naive results

Operators in $SL(2)$ sector

= components of $SL(2)$ irrep with $S = -1/2$

$$\left\{ z, D_+ z, \frac{1}{2!} D_+^2 z, \frac{1}{3!} D_+^3 z, \dots \right\}$$

$$\left\{ |0\rangle, a_+ |0\rangle, a_+^2 |0\rangle, a_+^3 |0\rangle, \dots \right\}$$

oscillator rep. $[a, a^\dagger] = 1$

$$\left\{ 1, z, z^2, z^3, \dots \right\}$$

differential rep. $[\partial_z, z] = 1$

differential realization of $SL(2)$ generators

$$\left\{ \begin{array}{l} S^- = -\partial_z \\ S^0 = z \partial_z + S \\ S^+ = z^2 \partial_z + 2S z \end{array} \right. \quad \begin{array}{l} S = -\frac{1}{2} \rightarrow SL(2) \\ S = \frac{1}{2} \mathbb{I} N \rightarrow SU(2) \end{array}$$

$$Q = \text{tr} (D_+^S Z^J) + \dots$$

$\leadsto$ spin chain with J sites

$\leadsto$ wave-fn $\psi(z_1, z_2, \dots, z_J)$

Twist-two case ($J=2$)

$$z_1^k z_2^{n-k} \sim (a_1^+)^k (a_2^+)^{n-k} |0\rangle \sim \frac{\text{tr}(D_+^k z D_+^{n-k} z)}{k! (n-k)!}$$

Compute one-loop operator mixing:

$$H_{12}(z_1^k z_2^{n-k}) = \sum_{k'=0}^n \left\{ \delta_{kk'} (h_k + h_{n-k}) + \frac{(1 - \delta_{kk'})}{|k - k'|} \right\} z_1^{k'} z_2^{n-k'}$$

$$h_n = \sum_{j=1}^n \frac{1}{j} = \psi(j+1) - \psi(1) : \text{harmonic number}$$

[Beisert, 0307015] [Belitzky, Derkachov, Korchemsky, Manashov, 031104]

Eigenstate of $H_{12} \leftrightarrow SL(2)$ spin- j representation

$$|j\rangle \sim (z_1 - z_2)^j = \sum_{l=0}^j (-1)^l \binom{j}{l} z_1^{j-l} z_2^l$$

In fact, one can show:

$$H_{12} (z_1 - z_2)^j = 2h(j) (z_1 - z_2)^j$$

$$\Rightarrow H_{12} = \sum_{j=0}^{\infty} 2h(j) \underline{P_{12,j}}$$

projector to spin- j rep

Hamiltonian of $XXZ_{-1/2}$ with length L

$$H = \sum_{k=1}^L H_{k \rightarrow k+1} = 2 \sum_k \left\{ \psi(\hat{J}_{k,k+1}) - \psi(1) \right\}$$

$\hat{J}_{12}$ is the "square root" of Casimir

$$S_{12}^A = S_1^A + S_2^A, \quad C_{12} = S_{12}^0(S_{12}^0 - 1) + S_{12}^+ S_{12}^-$$

$$\text{such that } C_{12} |j\rangle = j(j-1) |j\rangle$$

$$\Rightarrow \text{formally define } \hat{J}_{12} \text{ by } C_{12} \equiv \hat{J}_{12}(\hat{J}_{12} - 1)$$

Solving $XXZ_{-1/2}$ Bethe Ansatz

$$\left(\frac{\nu + i/2}{\nu - i/2} \right)^J \prod_{k=1}^S \frac{\nu - u_k + i}{\nu - u_k - i} = -1 \quad (k=1, 2, \dots, S)$$

for $Q \sim \text{tr}(D_+^S z^J)$

Twist-two ($J=2$)

there is only one HWS at each spin S

Hamiltonian is diagonal: $H = H_{12} = 2h(S)$

Position of all Bethe roots is known:

$$Q(u) = {}_3F_2 \left(-S, S+1, \frac{1}{2} + iu ; 1, 1 ; 1 \right)$$

No "finite" bound states in $SL(2)$ sector

Outermost roots scale as $u_j \sim \mathcal{O}(S)$

$$\leadsto \text{energy}; \quad \gamma = \frac{\lambda}{8\pi^2} \sum_{j=1}^S \frac{1}{u_j^2 + 1/4} = \frac{\lambda}{2\pi^2} h(S)$$

TQ-relation:

$$t(u) Q(u) = Q(u+i) \left(u + \frac{i}{2}\right)^J + Q(u-i) \left(u - \frac{i}{2}\right)^J$$

$$t(u) = 2u^2 - S(S+1) - 1/2 \quad \text{from symmetry}$$

§ Large Spin Limit

$$h(S) \rightarrow \log S + \gamma_E + \mathcal{O}\left(\frac{1}{S}\right)$$

$$\Delta - S - 2 = f(\lambda) \log S + \dots$$

$$f(\lambda) = \frac{\lambda}{2\pi^2} + \mathcal{O}(\lambda^2) : \text{"scaling function"}$$

{ $\log S$ behaviour agrees with string theory
No wrapping corrections at the leading order
related to the vev of Wilson loop with cusp

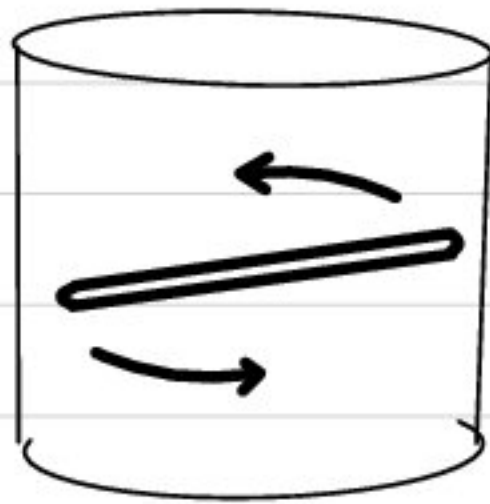
GKP folded string on AdS_3 :

$$ds^2 = -\cosh^2 \rho dt^2 + d\rho^2 + \sinh^2 \rho d\phi^2$$

can solve
Virasoro
without S^5

Ansatz: $t = k\tau$, $\phi = \omega\tau$, $\rho = \rho(\sigma)$

Elliptic profile:



ρ_0 : turning point

$\rho_0 \rightarrow \infty$ if $\omega \rightarrow k$

$$E \sim S \sim \sqrt{\lambda} e^{2\rho_0}, \quad E-S \sim \frac{\sqrt{\lambda}}{2\pi} 4\rho_0 \sim \frac{\sqrt{\lambda}}{\pi} \log S$$

$\therefore$) worldsheet has 4 segments $\rho \in [0, \rho_0]$

Therefore

$$E - S = f(\lambda) \log S + \dots$$

$$f(\lambda) = \frac{\sqrt{\lambda}}{\pi} + \dots \quad (\lambda \gg 1)$$

The scaling function is related to the anomalous dimension of cusped Wilson loop

$$\left\{ \begin{array}{l} W_{ren} = Z W_{bare} \sim \left(\frac{\Lambda_{UV}}{\Lambda_{IR}} \right)^{-\Gamma_{cusp}(\phi, \theta)} \\ \Gamma_{cusp} = - \frac{d \log Z}{d \log \mu} \end{array} \right.$$

Maldacena Wilson loop (in Euclidean space)

$$W = P \exp \oint ds \left(i A_\mu \frac{dz^\mu}{ds} + |\dot{x}^\mu| \phi^I n^I \right)$$

One-loop renormalization



A Feynman diagram representing a vertex. Two external lines, one incoming from the top-left and one outgoing to the bottom-right, meet at a central vertex. A wavy line (representing a scalar field) connects this vertex to another vertex above it. The incoming line is labeled with the vector $(\dot{x}_1^\mu, n_1^I)$ and the outgoing line with $(\dot{x}_2^\mu, n_2^I)$.

$$\propto (\dot{x}_1 \cdot \dot{x}_2 - n_1 \cdot n_2) = (\cos \phi - \cos \theta)$$

Take the limit $\phi \equiv i\gamma \rightarrow i\infty$

$$x^\mu = (s \cosh \gamma, s \sinh \gamma, 0, 0) \rightarrow (s e^\gamma, s e^\gamma, 0, 0)$$

lightlike limit

Weak coupling result :

[Drukker, Gross, Ooguri, 9904191] [Drukker, Fenn, 1105.5144]

$$\langle W_{\text{cusp}} \rangle \sim \exp \left(-T \sum_n \left(\frac{\lambda}{16\pi^2} \right)^n \Gamma_{\text{cusp}}^{(n)} \right)$$

(used conformal map to send cusp to spatial infinity)

$$\Gamma_{\text{cusp}}^{(1)} = 2 \frac{\cos \phi - \cos \theta}{\sin \phi} \phi \rightarrow 2\phi \quad (\phi \rightarrow i\infty)$$

suggesting $\Gamma_{\text{cusp}}(i\gamma, \theta) \rightarrow \gamma f(\theta) \quad (\gamma \rightarrow \infty)$

Higher loop :
$$\gamma = \sum_{n=1}^{\infty} \left(\frac{\lambda}{16\pi^2} \right)^n \gamma_n$$

$$\gamma_1 = 8 S_1$$

$$\gamma_2 = -16 \left\{ S_3 + S_{-3} - 2 S_{-2,1} + 2 S_1 (S_2 + S_{-2}) \right\}$$

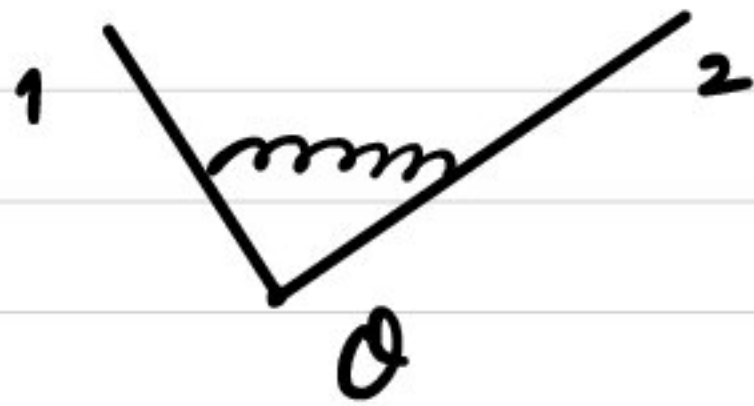
$$S_\phi = 1, \quad S_{a_1 \dots a_n} = \sum_{k=1}^S \frac{(\text{sign } a_1)^k}{k |a_1|} S_{a_2 \dots a_n}$$

All terms of γ_n have the same transcendentality

$$\text{wt}(S_{a_1 a_2 \dots}) = |a_1| + |a_2| + \dots$$

related to max. transcendental terms in QCD

Why Cusp dimension = scaling function ? (HARD)



cusp $\sim$ insertion of detect op.

lightlike limit $\Rightarrow \mathcal{O} \sim D_+^S \mathcal{Z}$

One-loop in momentum space $\sim \int d^D k \frac{1}{(\dot{x}_1 \cdot k)(\dot{x}_2 \cdot k)}$

divergent if collinear; $k^\mu \parallel \dot{x}_1^\mu$ or $\dot{x}_2^\mu$

collinear gluons are the main contribution for $\Gamma_{\text{cusp}}, f(g)$

$\Rightarrow$ should be true for any gauge theory, finite N

§ BFKL limit

$$\mathcal{Q} = \text{tr}(\mathbf{z} \overleftrightarrow{D}_+^s \mathbf{z}) \Rightarrow \gamma_1 = \frac{\lambda}{2\pi^2} h(s)$$

γ_1 diverges if $s \in \{-1, -2, \dots\}$, Expand around $s = -1$

$$\gamma_1 = \frac{\lambda}{2\pi^2} h(-1+\omega) = \frac{\lambda}{2\pi^2} \left(-\frac{1}{\omega} + \mathcal{O}(\omega^0) \right)$$

BFKL equation:

[Kotikov, Lipatov
hep-ph/0208220]

$$\omega = \frac{\lambda}{4\pi^2} \left\{ 2\psi(1) - \psi\left(\frac{1+\Delta}{2}\right) - \psi\left(\frac{1-\Delta}{2}\right) \right\}$$

$$= \frac{\lambda}{2\pi^2} \left(-\frac{1}{\Delta-1} + \dots \right) \rightarrow \text{agreement} \\ \text{if } \Delta = 1 + \gamma_1$$

ω_{BFKL} is the "singularity" of conformal 4pt function

[Costa, Gonçalves, Penedones, 1209.4355]

$$\langle \varnothing \varnothing \varnothing \varnothing \rangle \sim g(u, v) \xrightarrow{\text{Mellin}} M(s, t)$$

$$M(s, t) = \sum_n \int d\nu \, b_{n, \nu} \, \underline{M_{n, \nu}(s, t)}$$

partial wave for $SL(2, \mathbb{C})$

replace $\sum_n \rightarrow \int dJ$; $b(J, \nu) \sim \frac{b_0}{J - J_*$

leading contribution in the Regge limit ($s \rightarrow \infty$, t fixed)

The pole position $J_* = J_*(\omega_{\text{BFKL}})$ is given by
"summing over all ladder diagrams"



$\leadsto$ BFKL equation

Why related to $\gamma(S)$ around $S = -1$?

- Singularity at the same quantum number & kinematics
- consistent with integrability predictions (all orders in λ)
- perhaps "trivial relation" in QED, no ladder sums

Historically, Regge limit of QCD $\rightarrow$ BFKL equations

$$\omega \Phi(k) \sim \int d^2k \ K_{\text{BFKL}}(k, k') \Phi(k')$$

$\leadsto$ H_{12} of integrable $SL(2, \mathbb{C})$ spin chain, $H = \sum_k H_{k, k+1}$

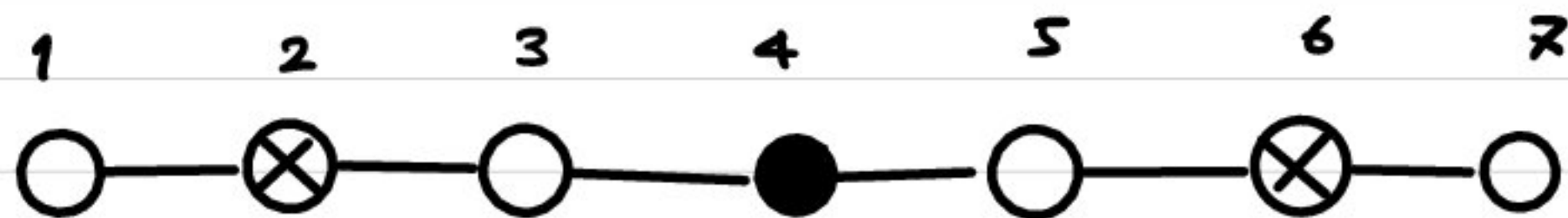
4D high energy scattering

transverse 2D plane becomes "simple"

$\left\{ \begin{array}{ll} \text{"BFKL Pomeron" for QCD} & \leadsto N=4 \text{ SYM ?} \\ \text{"Soft Pomeron" for hadrons} & \leftarrow \text{real experiment} \end{array} \right.$

{ Asymptotic Bethe Ansatz

One-loop Bethe Ansatz for $psu(2,2|4)$



bosonic fermionic

momentum-carrying

$V = [0, 0, 0, 1, 0, 0, 0]$: Dynkin label

example : 

vector rep. for $so(6)$

$V = [0, 1, 0]$

Cartan Matrix

$$M = \left(\begin{array}{cc|cc|cc} -2 & 1 & & & & \\ 1 & 0 & -1 & & & \\ \hline & -1 & 2 & -1 & & \\ & & -1 & 2 & -1 & \\ & & & -1 & 2 & -1 \\ \hline & & & & -1 & 0 & 1 \\ & & & & & 1 & -2 \end{array} \right) \begin{array}{l} B \\ F \\ B \\ B \\ B \\ F \\ B \end{array}$$

Called "Beauty" diagram

Dynkin diagram of super Lie alg is not unique

Bethe Ansatz

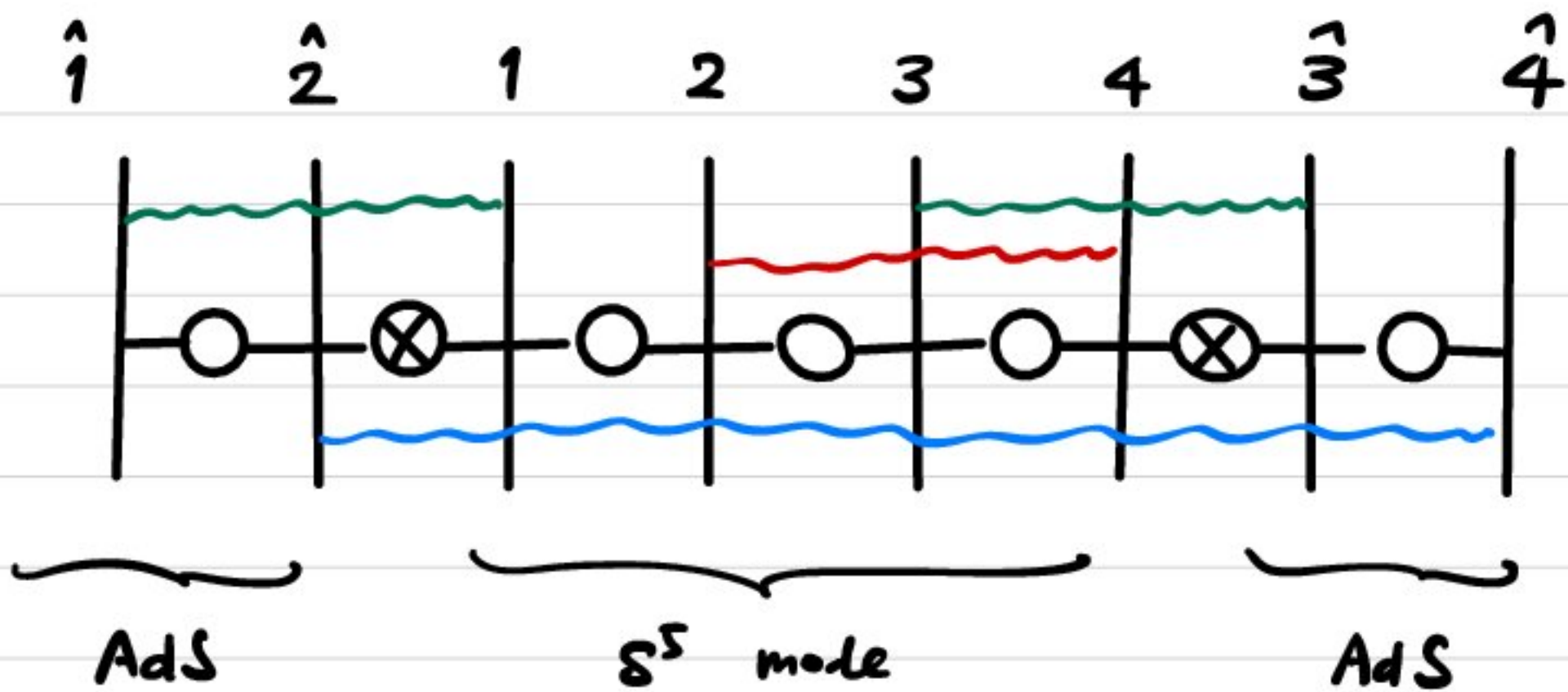
$$\left(\frac{u_j + \frac{i}{2} V_{kj}}{u_j - \frac{i}{2} V_{kj}} \right)^L = \prod_{l \neq j} \frac{u_j - u_l + \frac{i}{2} M_{kj,kl}}{u_j - u_l - \frac{i}{2} M_{kj,kl}}$$

Zero-momentum

$$\prod_j \frac{u_j + \frac{i}{2} V_{kj}}{u_j - \frac{i}{2} V_{kj}} = 1$$

Energy

$$E = \sum_j \left(\frac{i}{u_j + \frac{i}{2} V_{kj}} - \frac{1}{u_j - \frac{i}{2} V_{kj}} \right)$$



S^1 , J^1 J^2 J^3 , S^2 : Cartan

$$\left\{ \begin{array}{ccc} \hat{1}\hat{2} & \hat{1}\hat{3} & \hat{1}\hat{4} \\ \hat{3}\hat{4} & \hat{2}\hat{4} & \hat{2}\hat{3} \end{array} \right\}$$

$$\left\{ \begin{array}{ccc} 12 & 13 & 14 \\ 34 & 24 & 23 \end{array} \right\}$$

$$\left\{ \begin{array}{c} \hat{1}a, \hat{2}a \\ a\hat{3}, a\hat{4} \end{array} \right\} \quad a=1,2,3,4$$

$$\vec{y} \cdot \vec{y} = -1$$

AdS₅

$$\vec{x} \cdot \vec{x} = 1$$

S⁵

16 fermions

Assuming that all roots are regular
(= HWS of $\text{PSu}(2,2|4)$)

Number of Bethe roots & length
($k^1, k^2, \dots, k^3, L$)

$\longleftrightarrow$ Cartan charges of $\text{PSu}(2,2|4)$
($\Delta, S^1, S^2, J^1, J^2, J^3$)

Also "duality transformation" can change vacuum

$$|0\rangle = \text{tr } Z^L \longrightarrow |0\rangle = \text{tr } F^L, \quad F = F_{1+i2, 3+i4}$$

3-loop in $SU(2)$ sector:

[Beisert, Dippel, Staudacher, 0405001]

[Rej, Serban, Staudacher, 0512077]

higher loop $\rightarrow$ long-range interaction

Hubbard model:

$$n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$$

$$H = -t \sum_i \sum_{\sigma=\uparrow, \downarrow} \left(\underbrace{c_{i,\sigma}^\dagger c_{i+1,\sigma}}_{\text{hopping}} + \underbrace{c_{i+1,\sigma}^\dagger c_{i,\sigma}}_{\text{Coulomb}} \right) + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

Four states $\{ \phi, \uparrow, \downarrow, \uparrow\downarrow \}$

Z, W on anti-ferro vacuum

$U \rightarrow \infty$, expand hopping term

- Reproduces dilatation in $SU(2)$ sector up to 3 loops
- Half-filling $\Rightarrow$ (dual) Lieb-Wu equations
- identify W as bound state of $(0, \uparrow)$

$$\Rightarrow e^{ip_n L} = \left(\frac{x(u_n + i/2)}{x(u_n - i/2)} \right)^L = \prod_{j \neq n}^n \frac{u_n - u_j + i}{u_n - u_j - i}$$

$$u(p) = x + \frac{1}{x} = \frac{1}{2} \cot \frac{p}{2} \sqrt{1 + \frac{\lambda}{\pi^2} \sin^2 \frac{p}{2}}$$

$$\gamma = \frac{i\lambda}{8\pi^2} \sum_j \left\{ \frac{1}{x(u_j + i/2)} - \frac{1}{x(u_j - i/2)} \right\}$$

x is called Zhukovsky variable

$$x + \frac{1}{x} = u \Leftrightarrow x = \frac{1}{2} \left(u + \sqrt{u^2 - 4} \right)$$

$$x = \pm 1 \Leftrightarrow u = \pm 2 : \text{branch points}$$

"stung" branch choice is $|x(u)| \geq 1$ for $\forall u \in \mathbb{C}$

$$x^{\pm} \equiv x(u \pm \frac{i}{2}) = e^{\pm i p/2} \frac{1 + \sqrt{1 + \frac{\lambda}{\pi^2} \sin^2 \frac{p}{2}}}{\sqrt{\frac{\lambda}{\pi}} \sin \frac{p}{2}}$$

$\rightarrow 1$ if $\lambda \gg 1$

4-loop (& strong coupling) $\Rightarrow$ Hubbard BAE incorrect

Two sources of corrections:

1) Dressing phase, (RHS) = $\frac{u_n - u_j + i}{u_n - u_j - i} e^{2i\theta(u_n, u_j)}$

2) Wrapping corrections, (Energy) = $E_{BAE} + \delta E$

"Asymptotic Bethe Ansatz" takes care of (1)

Beisert-Staudacher eq. with BES phase

[0504190, 0610251]

$$\left(\frac{x_{4,k}^+}{x_{4,k}^-} \right)^L = \prod_{j \neq k}^{K_4} \frac{x_{4,k}^+ - x_{4,j}^-}{x_{4,k}^- - x_{4,j}^+} \frac{1 - \frac{1}{x_{4,k}^+ x_{4,j}^-}}{1 - \frac{1}{x_{4,k}^- x_{4,j}^+}} \frac{\sigma^2(x_{4,k}, x_{4,j})}{\text{dressing}}$$

$$\times \prod_{j=1}^{K_3} \frac{x_{4,k}^- - y_{3,j}}{x_{4,k}^+ - y_{3,j}} \prod_{j=1}^{K_5} \frac{x_{4,k}^- - y_{5,j}}{x_{4,k}^+ - y_{5,j}} \left(\sim \frac{u_k - u_j + i}{u_k - u_j - i} \right)$$

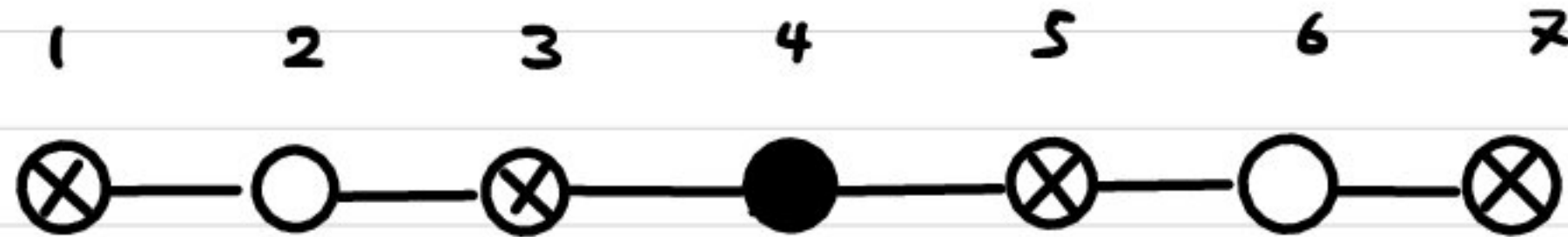
$$\times \prod_{j=1}^{K_1} \frac{1 - \frac{1}{x_{4,k}^- y_{1,j}}}{1 - \frac{1}{x_{4,k}^+ y_{1,j}}} \prod_{j=1}^{K_7} \frac{1 - \frac{1}{x_{4,k}^- y_{7,j}}}{1 - \frac{1}{x_{4,k}^+ y_{7,j}}}$$

1, 3, 5, 7

fermionic roots

$y \leftrightarrow 1/y$

PSU(2,2|4) Dynkin:



$$V = (0, 0, 0, 1, 0, 0, 0)$$

6 more equations for auxiliary roots

$$y_1, \alpha_2^\pm, y_3, \quad y_5, \alpha_6^\pm, y_7$$

not "beauty" Dynkin:

fermionic dualities with Zhukovsky var. is complicated

At one-loop we know everything:

Dilatation operator, wave-fn, Factorized S-matrix

At higher loops:

$D \rightarrow$ Possible to compute but tedious
Wrapping corrections from 4-loops

$\psi \rightarrow$ Length-changing effect: $\psi\psi \sim XYZ$

S-matrix $\rightarrow$ still simple $\rightarrow$ BAE!

N.B. We don't know the operator D we're diagonalizing
just its eigenvalues are conjectured.